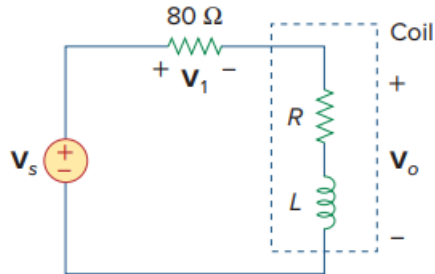


Problem

An industrial coil is modeled as a series combination of an inductance L and resistance R , as shown in Fig. 9.90. Since an ac voltmeter measures only the magnitude of a sinusoid, the following measurements are taken at 60 Hz when the circuit operates in the steady state:

$$|V_s| = 145 \text{ V}, \quad |V_1| = 50 \text{ V}, \quad |V_o| = 110 \text{ V}$$

Use these measurements to determine the values of L and R .



Step-by-step solution

Step 1 of 4

The inductive reactance is,

$$\begin{aligned} X_L &= 2\pi fL \\ &= 2\pi(60)L \\ &= 377L \end{aligned}$$

The current in the circuit is,

$$\begin{aligned} \mathbf{I} &= \frac{\mathbf{V}_s}{80 + R + jX_L} \\ &= \frac{145}{80 + R + jX_L} \end{aligned}$$

The voltage across the 80Ω resistor is,

$$\begin{aligned} \mathbf{V}_1 &= 80\mathbf{I} \\ &= (80) \left(\frac{145}{80 + R + jX_L} \right) \\ &= \frac{11600}{80 + R + jX_L} \end{aligned}$$

$$|V_1| = \frac{11600}{\sqrt{(80+R)^2 + X_L^2}} \dots\dots (1)$$

Step 2 of 4

The output voltage is,

$$\mathbf{V}_o = \left(\frac{R + jX_L}{80 + R + jX_L} \right) (145 \text{ V})$$

$$|\mathbf{V}_o| = \frac{\sqrt{R^2 + X_L^2}}{\sqrt{(80 + R)^2 + X_L^2}} (145 \text{ V}) \dots\dots (2)$$

Divide equation (1) with equation (2).

$$\frac{|\mathbf{V}_1|}{|\mathbf{V}_o|} = \frac{\frac{11600}{\sqrt{(80 + R)^2 + X_L^2}}}{\frac{\sqrt{R^2 + X_L^2}}{\sqrt{(80 + R)^2 + X_L^2}}} (145 \text{ V})$$

Substitute 50 V for $|\mathbf{V}_1|$ and 110 V for $|\mathbf{V}_o|$ in the equation.

$$\frac{50}{110} = \frac{11600}{(145 \text{ V}) \sqrt{R^2 + X_L^2}}$$

$$\sqrt{R^2 + X_L^2} = \frac{11600}{(0.4545)(145 \text{ V})}$$

$$R^2 + X_L^2 = (176)^2$$

$$R^2 + X_L^2 = 30976 \dots\dots (3)$$

Step 3 of 4

Recall equation (1).

$$|\mathbf{V}_1| = \frac{11600}{\sqrt{(80 + R)^2 + X_L^2}}$$

Substitute 50 V for $|\mathbf{V}_1|$ in the equation.

$$50 = \frac{11600}{\sqrt{(80 + R)^2 + X_L^2}}$$

$$\sqrt{(80 + R)^2 + X_L^2} = 232$$

$$(80 + R)^2 + X_L^2 = 53824$$

$$6400 + 160R + R^2 + X_L^2 = 53824$$

Substitute 30976 for $R^2 + X_L^2$ in the equation.

$$6400 + 160R + 30976 = 53824$$

$$160R = 16448$$

$$R = 103 \, \Omega$$

Therefore, the value of resistance, R is $103 \, \Omega$.

Step 4 of 4

Recall equation (3).

$$R^2 + X_L^2 = 30976$$

Substitute $103\ \Omega$ for R in the equation.

$$(103)^2 + X_L^2 = 30976$$

$$10609 + X_L^2 = 30976$$

$$X_L^2 = 20367$$

$$X_L = 143\ \Omega$$

Substitute $377L$ for X_L in the equation.

$$377L = 143\ \Omega$$

$$L = 379\ \text{mH}$$

Therefore, the value of inductor, L is 379 mH.